

ABHINAV TARIGOPPULA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Leetcode](#) | [ResearchGate](#)

SUMMARY

Final-year B.Tech CSE (AI & ML) student at GITAM University with industry experience spanning machine learning, generative AI, computer vision, and full-stack deployment. Currently building defect-detection pipelines at APEPDCL and enrolled in PwC Launchpad, with 12 merged PRs into open-source frameworks including unsloth, HuggingFace, and pydantic. Research paper selected for IEEE ISED 2026 at NIT Warangal; co-authored additional work published on ResearchGate. Open to full-time roles, PPO, and research collaborations.

EDUCATION

GITAM University, Visakhapatnam, Andhra Pradesh

Aug 2023 – May 2027

B.Tech, Computer Science & Engineering (AI/ML Specialization) · CGPA: 8.12

Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, DBMS, Operating Systems, Computer Networks

FIITJEE, Visakhapatnam

2020 – 2022

Class XII — Mathematics, Physics, Chemistry (MPC)

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++

ML & AI: PyTorch, TensorFlow, Scikit-learn, Computer Vision, Object Detection, YOLOv8, LangChain, OpenAI API, Hugging Face Transformers, NLP, RAG, LLM Fine-Tuning, Prompt Engineering, Kubernetes

Software Engineering: FastAPI, Node.js, REST APIs, Vector Databases (FAISS, Pinecone), asyncio, pytest, AST Parsing, Docker

Tools & DevOps: Git, GitHub Actions, CI/CD, AWS, Linux, React.js, Next.js

CS Foundations: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

AI/ML Intern — Computer Vision — APEPDCL (Eastern Power Distribution Co. of AP)

On-site · Jun 2026 – Present

- Building YOLOv8 + MiDaS computer-vision pipeline to detect power-infrastructure defects and estimate DTR pole heights from field imagery for a state utility serving millions of customers.
- Constructing end-to-end CV data pipelines covering image capture, annotation (classification, detection, segmentation), and model training/evaluation on real-world utility datasets using PyTorch/TensorFlow.

PwC Launchpad Enterprise Applications Salesforce — PwC India

Remote · Feb 2026 – Present

- Ranked among top performers in PwC's flagship GenAI & Data program; achieved Level 10 (1,500+ XP) out of all cohort participants.
- Developed LLM prompt workflows and enterprise data architecture designs applied to real client advisory scenarios.

Machine Learning & AI Intern — OneStop AI

Remote · May 2025 – Aug 2025

- Boosted classification accuracy by ~12% (78%→90%) across 3 production ML models via hyperparameter tuning and neural architecture search.
- Cut inference latency by ~18% (220ms→180ms) by refactoring preprocessing pipelines and applying model quantization.
- Engineered end-to-end data pipelines (preprocessing, feature engineering, batch inference) with CI/CD integration for repeatable, auditable model delivery.

AI/ML Engineering Intern — AVAIntern

On-site · Visakhapatnam · May 2026 – Jun 2026

- Developed and evaluated 5+ ML models on datasets of 10,000+ records, achieving up to 92% prediction accuracy through feature engineering and hyperparameter tuning.
- Automated data preprocessing and analysis workflows using Python, Pandas, and Scikit-learn, reducing manual data preparation time by 40%.
- Performed EDA and model evaluation across 15+ features, improving model performance by 18%; communicated findings to stakeholders via interactive visualizations.

PROJECTS

MarkMe — Face Recognition Attendance System [Live Demo ↗](#)

Python, OpenCV, face_recognition, FastAPI, React ·

- Built and deployed a real-time face-recognition attendance system achieving 94.3% accuracy on a 500+ student dataset; actively used at GITAM University, eliminating manual attendance.
- Engineered a FastAPI backend with async processing supporting concurrent recognition streams and a React dashboard for attendance analytics.

Subham Digitals — Family Business Website · [Website](#)

- Designed and deployed a production website for a family-run luxury wedding photography studio (est. 2006, Visakhapatnam) — live, actively used for client bookings
- Built a multi-category photo gallery across 8+ wedding story types (candid, church, nikah, haldi, outdoor, portraits) with smooth navigation and mobile-responsive layout
- Implemented SEO metadata, Open Graph tags, and Twitter cards for discoverability; integrated WhatsApp direct booking and consultation flow for client conversions.

PathForge — AI Placement Platform [Live Demo ↗](#)

Python, FastAPI, LangChain, OpenAI API, FAISS/Pinecone, React

- Designed and shipped a full-stack AI platform generating role-specific study material and mock interviews via LLM pipelines; owned end-to-end from backend APIs to production React UI.
- Architected a RAG pipeline (embedding → retrieval → prompt assembly) over FAISS/Pinecone achieving ~30% improvement in answer relevance during evaluation.

CodeHound — Python Static Analyzer [GitHub ↗](#)

Python, ast, pytest, GitHub Actions · ~750 LOC, zero dependencies

- Built an AST-based static analysis tool detecting async-safety and correctness bugs (blocking calls, fire-and-forget tasks, mutable defaults, resource leaks) — bugs it surfaced were merged into unsloth (40k★) and agno (25k★).

OPEN SOURCE CONTRIBUTIONS

- 12 PRs merged across 20+ organizations — unsloth (40k★), HuggingFace (huggingface_hub + accelerate + peft), agno (25k★), mem0 (35k★), pydantic (22k★), marimo (11k★ YC), xorbisai/inference — covering async event-loop blocking fixes, fire-and-forget task bugs, mutable-default (B006) fixes, resource leaks, and a new public-API feature; all reviewed and merged by core maintainers.
- Code shipped in huggingface_hub v1.17.0, live on PyPI. Bug classes fixed: blocking calls in async (time.sleep, requests.get), fire-and-forget tasks, mutable defaults, unclosed file handles, and missing documentation on public APIs.
- Active PRs under review at vLLM, Microsoft autogen, OpenAI Agents SDK, future-agi, and sglang — same async-safety bug class, different codebases.

CERTIFICATIONS & RESEARCH

- **micro1** — AI & Data Advisory (May 2026) · Passed AI-proctored technical interview, certified by micro1
- **Infosys SpringBoard** – Database and SQL
- **IBM SkillsBuild** — Code Generation & Optimization Using IBM Granite (Jun 2026, Credly-verified)
- **Cisco Networking Academy** — Introduction to Modern AI (May 2026)
- **FutureSkills Prime / Nasscom** — Digital Application Fundamentals, STEM (May 2026, Govt. of India / MeitY-backed)
- **AWS Forage** — Solutions Architecture Job Simulation: designed scalable hosting architecture on AWS (Jul 2025)
- **TCS iON Career Edge** — Young Professional: Communication, Interview Skills, Business Etiquette (Jun 2025)
- **Udemy** — Mastering AI Agents Bootcamp: Build Smart Chatbots & Tools (Jun 2025)
- Co-authored research paper: "ML-Based Student Performance Prediction: A Comparative Study Using Ensemble Methods and XAI" — submitted to ISED-2026 at NIT Warangal (1st author); additional target venues: AISP-2026, IEEE Engineering Informatics 2026.

ACHIEVEMENTS

- Solved 200+ DSA problems on LeetCode (Easy, Medium, Hard) — active placement prep across arrays, trees, graphs, and DP.
- GitHub badges: Pull Shark (Silver), YOLO, Galaxy Brain — awarded for merged PRs and significant community contributions.
- Ranked Top 5 at the AI & Machine Learning Hackathon hosted by Techgyan Technologies at BITS Pilani, Hyderabad — 24-hour in-person competition.
- Participant, Smart India Hackathon — India's largest national hackathon with 1M+ student participants.
- Active member of the college fitness community; maintain a consistent early-morning training routine.
- Co-manage digital operations for family-run wedding photography studio (est. 2006) — handling digital marketing, videography/photography shoots, client data management, and online delivery